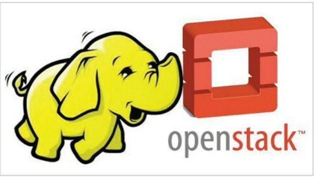

Figure 1: Hadoop on OpenStack

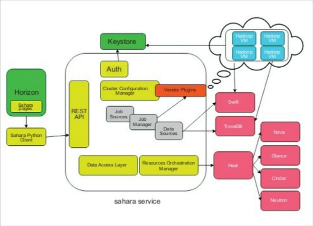

Figure 2: OpenStack Sahara architecture

Hadoop is an industry standard framework for storing and analysing a large data set with its fault tolerant Hadoop Distributed File System and MapReduce implementation. Scalability is a very common problem in a typical Hadoop cluster. OpenStack has introduced a project called Sahara –Data Processing as a Service. OpenStack Sahara aims to provision and manage data processing frameworks such as Hadoop MapReduce, Spark and Storm in a cluster topology. This project is similar to the data analytics platform provided by the Amazon Elastic MapReduce (EMR) service. Sahara deploys the cluster in a few minutes. Besides, OpenStack Sahara can scale the cluster by adding or removing worker nodes, based on demand.

The benefits of managing a Hadoop cluster with OpenStack Sahara are:

- Clusters can be provisioned faster and are easy to configure.
- Like other OpenStack services, the Sahara service can be managed through its powerful REST API, CLI and horizon dashboard.
- Plugins are available to support multiple Hadoop vendors such as Vanilla (Apache Hadoop), HDP (Ambari), CDH (Cloudera), MapR, Spark and Storm.
- Cluster size can be scaled up and down, based on demand.
- Clusters can be integrated with OpenStack Swift to store the data processed by Hadoop and Spark.
- Cluster monitoring can be made simple.

Apart from cluster provisioning, Sahara can be used as 'Analytics as a Service' for ad hoc or bursty analytic workloads. Here, we can select the framework and load the job data. It behaves as a transient cluster and will be automatically terminated after job completion.

Architecture

OpenStack Sahara is designed to leverage the core services and other fully managed services of OpenStack. It makes Sahara more reliable and capable of managing the Hadoop cluster efficiently. We can optionally use services such as Trove and Swift in our deployment. Let us look into the internals of the Sahara service.

- The Sahara service has an API server which responds to HTTP requests from the end user and interacts with other OpenStack services to perform its functions.
- Keystone (Identity as a Service) authenticates users and provides security tokens that are used to work with OpenStack, limiting users' abilities in Sahara to their OpenStack privileges.
- Heat (Orchestration as a Service) is used to provision and orchestrate the deployment of data processing clusters.
- Glance (Virtual Machine Image as a Service) stores VM images with operating system and pre-installed Hadoop/ Spark software packages to create a data processing cluster.
- Nova (Compute as a Service) provisions a virtual machine for data processing clusters.
- Ironic (Bare metal as a Service) provisions a bare metal node for data processing clusters.
- Neutron (Networking as a Service) facilitates networking services from basic to advanced topologies to access the data processing clusters.
- Cinder (Block Storage) provides a persistent storage media for cluster nodes.
- Swift (Object Storage) provides reliable storage to keep job binaries and the data processed by Hadoop/ Spark.
- Designate (DNS as a Service) provides a hosted zone to keep DNS records of the cluster instances. Hadoop services communicate with the cluster instances by their host names.
- Ceilometer (Telemetry as a Service) collects and stores the metrics about the cluster for metering and monitoring purposes.
- Manila (File Share as a Service) can be used to store job binaries and data created by the job.
- Barbican (Key Management Service) stores sensitive data such as password and private keys securely.
- Trove (Database as a Service) provides a database instance for the Hive metastore, and stores the states of the Hadoop services and other management services.